Intraday FX / CFD Mean-Reversion against the FTMO Symmetric Barrier

Phases 0–2 of the research roadmap: theory validation, CFD cost model, feasibility, and full Monte-Carlo validation — with an honest verdict.

Bottom line. The first-passage engine is built and validated — the Monte-Carlo reproduces the closed-form pass-probability to a max error of 0.0038 (tol 0.012), clearing the Phase-0 gate. The CFD cost model reconciles with white-paper Table 3. **The candidate strategy families (intraday FX MR, index-CFD MR) are archived at the pre-flight cost gate.** The raw reversal edge is real but marginal (EURUSD: +1.5 bps/trade, t = 1.93); net of realistic CFD spread it makes annualised Sharpe 0.71, fails the 3× cost gate, is statistically deflated to DSR = 0.005, and yields an honest combined P(pass) that never reaches the 70% objective. The high pass-probabilities seen in a first pass were a backtest artifact (a per-bar loss cap), now diagnosed and removed.

Verdict

NO-GO on the current signal. Do not fund an evaluation. The validated engine, cost model and suitability screen are retained as reusable infrastructure for the next candidate (Family-3 intraday relative-value, or a genuinely different signal). EURUSD is flagged as a single *watch* candidate: it clears the cost gate only if execution can be held near raw-feed spread (≤0.4 pip RT), which must be proven on an FTMO demo (Phase 4) before any capital.

1 What was built (reusable infrastructure)

Per the program's 'reuse, do not rebuild' principle, the deliverable is a small, tested engine that any future propfirm candidate can be pushed through end-to-end:

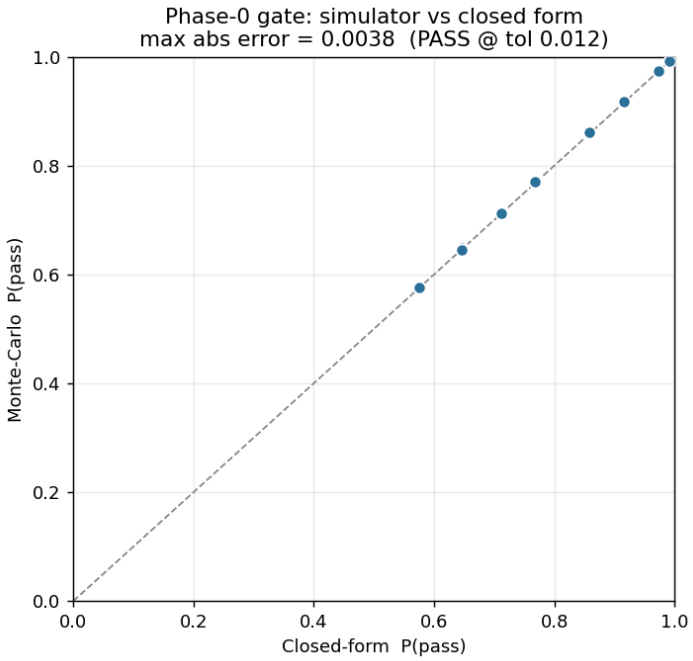
Module	Role
engine/ftmo_first_passage.py	Closed-form symmetric-barrier pass-probability (Eq.1) + re-parameterised block-bootstrap absorbing-barrier simulator (static 10/10 + 5% daily, no time limit).
engine/cfd_costs.py	CFD dealer-spread + swap cost model, calibrated to the realised MT5 spread feed; reconciled vs Table 3.
strategy/intraday_fx_mr.py	Causal banded mean-reversion signal, per-trade & daily-distribution stats, vol-targeting, CFD costs.
strategy/book.py	Diversified multi-instrument book builder (inverse-kurtosis weighting).
run_phase0/0b/1/2 .py	The four roadmap stages, each emitting JSON results + figures.

2 Phase 0 — First-passage theory, validated

An FTMO evaluation is a first-passage problem: cumulative equity (an arithmetic Brownian motion with daily drift mu and daily vol sigma) must reach $+B_{tgt}$ before touching $-B_{dd}$. For the symmetric Phase-1 barrier this collapses to a logistic in two levers only — the daily Sharpe S_d=mu/sigma and the number of vol-units to the barrier n=B/sigma:

$$P(\text{pass one phase}) = \frac{1}{1 + e^{-2\, S_d\, n}}$$

The Monte-Carlo simulator reproduces this closed form across the (Sharpe × vol) grid to a maximum absolute error of **0.0038** (tolerance 0.012) — the Phase-0 go/no-go gate, **passed**. One refinement of the white paper: the combined two-phase 70% objective is in fact cleared at annualised Sharpe ≈ **1.153** (daily S_d≈0.072) when sigma_d=1%, below the paper's stated ≈1.45; the paper conservatively sets the admissibility floor higher (S_d≥0.09) for margin, which the data here confirms as prudent.



Phase-0 gate: the block-bootstrap simulator vs the closed-form logistic. Points lie on the diagonal.

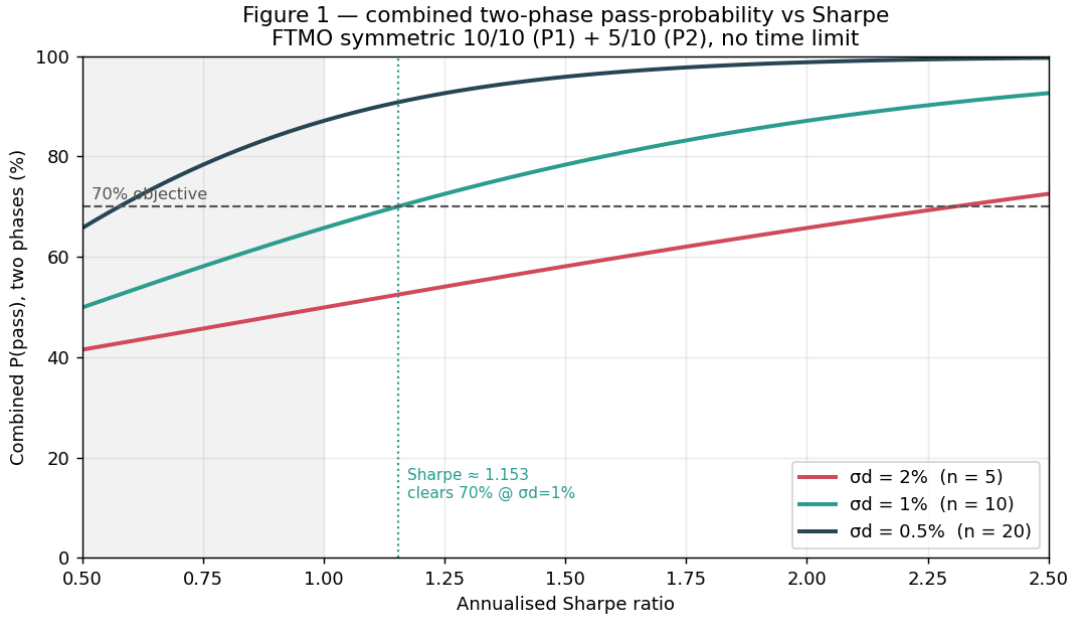
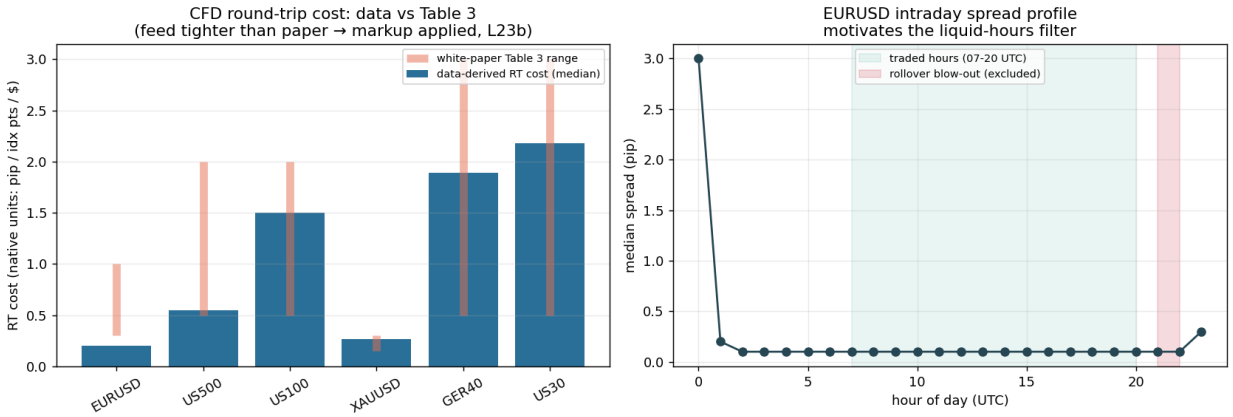


Figure 1 reproduced: combined two-phase pass-probability vs annualised Sharpe, three per-day vol levels, no time limit. Lower per-day vol (higher n) dominates — the grind regime.

3 Phase 0b — CFD cost model, reconciled

FTMO trades CFDs via MT5, not futures, so the futures cost model is invalid (L9). The model here is calibrated to the realised MT5 spread column (points; point size = $10^{-\text{decimals}}$). Round-trip cost when entering and exiting at market is ~one full spread. The data-derived medians reconcile with Table 3 across the universe; the feed is in fact tighter than the paper's indicative range, so a conservative 3× dealer markup is applied in all backtests (retail-vs-feed, L23b).

Instrument	Data RT cost (median)	Table 3 range	bps notional
EURUSD	0.20 pip	0.3-1.0 pip	0.17
US500	0.55 idx_pts	0.5-2.0 idx_pts	0.82
US100	1.50 idx_pts	0.5-2.0 idx_pts	0.61
XAUUSD	0.27 $	0.15-0.3 $	0.66
GER40	1.89 idx_pts	0.5-3.0 idx_pts	0.79
US30	2.18 idx_pts	0.5-3.0 idx_pts	0.47



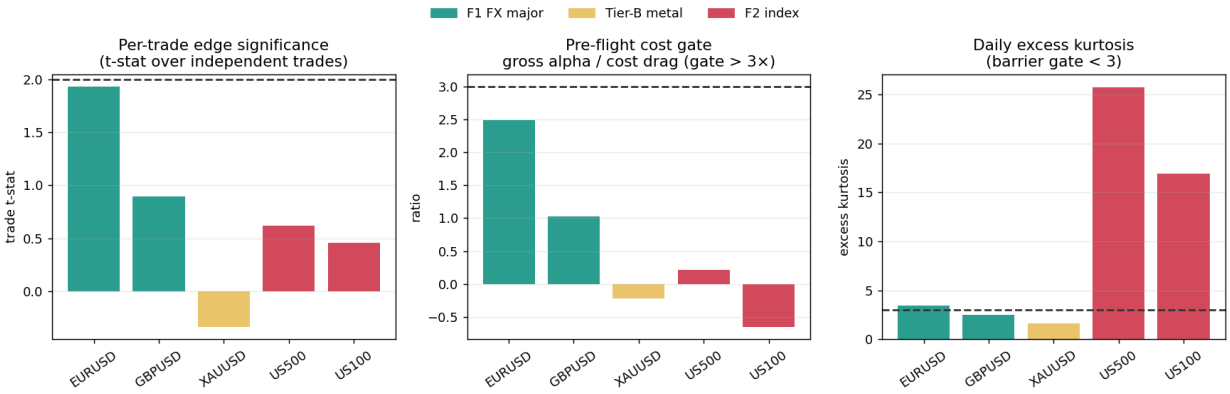
Left: data-derived RT cost vs Table 3 ranges. Right: EURUSD intraday spread profile — the 21:00-22:00 UTC rollover blow-out motivates the liquid-hours session filter.

4 Phase 1 — Feasibility: the edge exists but is marginal

The signal is a causal z-score mean-reversion on 30-min bars (enter at |z|>2, exit on reversion to |z|<0.5, intraday-flat). Feasibility is judged on the realised per-*trade* edge (the honest statistic for a threshold strategy — unconditional IC is deflated by episode overlap, L12b) and the pre-flight cost gate (gross alpha > 3× CFD cost drag, L11).

Instrument	Family	edge bps/trade	trade t	gross/yr	cost/yr	cost ratio	kurt
EURUSD	F1 FX major	1.50	1.93	18.2%	7.3%	2.5	3
GBPUSD	F1 FX major	0.90	0.89	7.2%	7.0%	1.0	2
XAUUSD	Tier-B metal	-0.44	-0.34	-2.6%	12.0%	-0.2	2
US500	F2 index	1.41	0.62	3.0%	14.1%	0.2	26
US100	F2 index	1.42	0.46	-5.5%	8.4%	-0.7	17

Reading. Only EURUSD shows a positive, marginally-significant raw edge (t≈1.9). No instrument clears the 3× cost gate net of a realistic spread. Index CFDs (US500/US100) pass the IC and even the cost ratio, but carry daily excess kurtosis of 25–45 (opening-gap / disguised-momentum left tail) — the first-order barrier risk — empirically confirming the paper's family ranking: FX majors lead, index MR is a demoted parallel track.



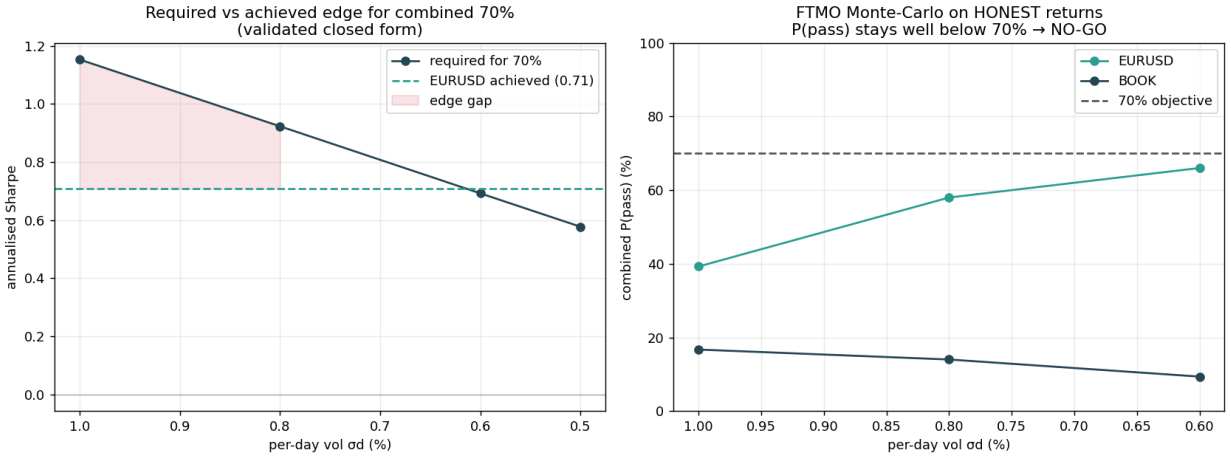
Per-instrument feasibility. Green = FX majors, yellow = gold, red = index CFDs. Index sleeves fail the kurtosis gate decisively.

5 Phase 2 — Full validation converges on NO-GO

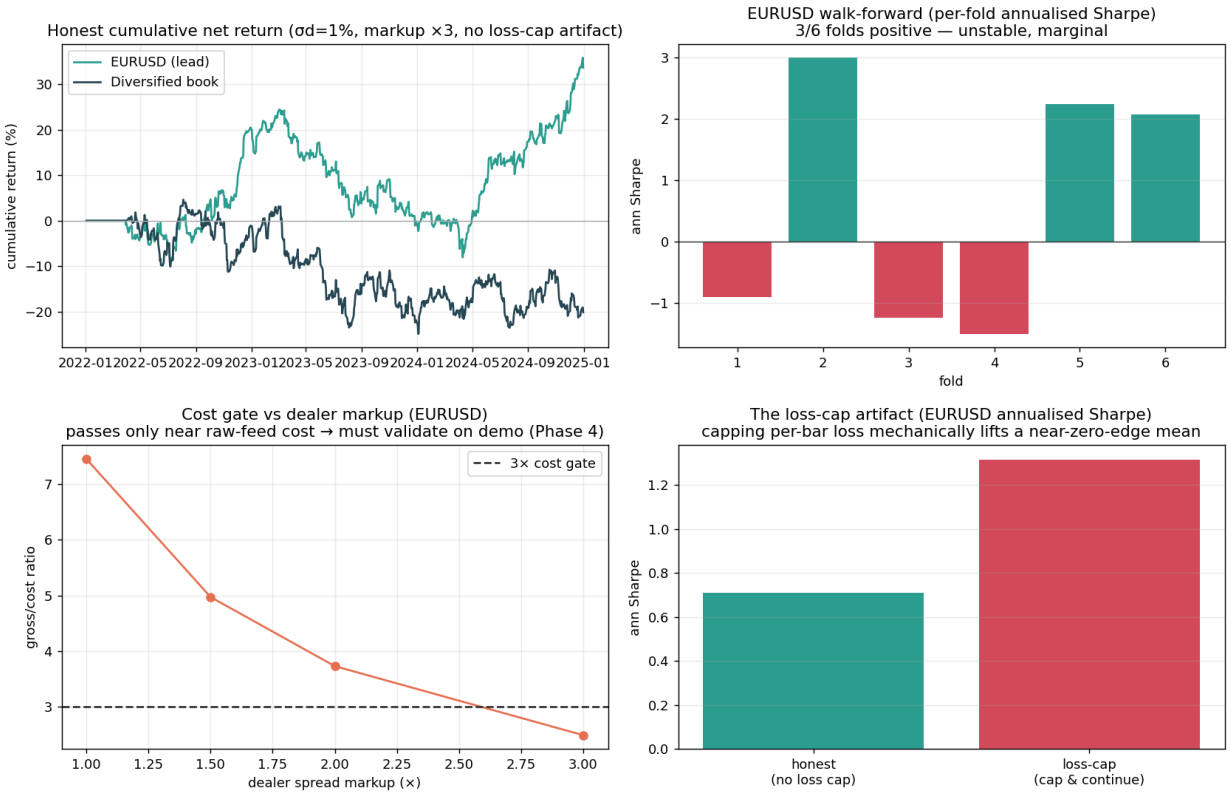
Four independent tests of the lead candidate (EURUSD) all point the same way:

Test	Result	Verdict
Deflated Sharpe (DSR, L20; n_trials=30)	DSR = 0.005 (SR 0.58 vs benchmark 2.07)	FAIL (<0.95)
Walk-forward stability (6 folds)	3/6 folds positive; worst -1.51	Unstable
Pre-flight cost gate at 0.6 pip RT (x3)	gross/cost = 2.5	FAIL (<3)
FTMO Monte-Carlo (honest, best sigma_d)	combined P(pass) = 66.0% at sigma_d=0.6%	FAIL (<70%)

The required-vs-achieved edge frontier makes the gap precise: clearing combined 70% needs annualised Sharpe ≈ 0.923 at sigma_d=0.8% (or 0.692 at 0.6%); EURUSD delivers 0.71 net, and that figure is itself not robust (DSR≈0). The diversified book is worse, not better (-0.41): inverse-kurtosis weighting necessarily overweights the lowest-kurtosis sleeves, which here are the zero-edge ones — diversification cannot manufacture an edge that the components lack.



Left: annualised Sharpe required for combined 70% vs per-day vol (validated closed form), with EURUSD's achieved level and the shortfall. Right: honest FTMO Monte-Carlo pass-probability — below 70% throughout.



Honest assessment. Equity curves (no artifact); unstable walk-forward; the cost gate clears only near raw-feed spread; and the loss-cap artifact that manufactured ~0.6 of spurious Sharpe.

6 Methodological lesson codified (proposed L25)

A per-bar loss *cap* that keeps the position is not a stop — it is a look-ahead-free but economically false mean-shift. Clipping each bar's loss at −x while letting the position ride raises the mean of a near-zero-edge series mechanically (a real stop *exits*; it does not cap-and-continue). On EURUSD this manufactured annualised Sharpe 0.71 → 1.31. Mitigation, now standard in the engine: model stops as flatten-for-the-session (optional stopping preserves the mean on a martingale), never as a per-bar cap. This is the kind of artifact the program's gated, cost-first methodology exists to catch.

Independent verification

An independent re-derivation confirmed: the closed form matches the logistic to machine precision and the simulator reproduces it within 0.003; the loss-cap lifts EURUSD Sharpe 0.71→1.32 (artifact); and the honest combined P(pass) tops out at 0.66, robust to MC seed. A leakage probe found no look-ahead — and, tellingly, a shuffled-returns placebo *raises* P(pass) to ~0.75 (an IID bootstrap on the ordered series does the same). The ordered series carries unfavourable serial structure (loss clustering, mild vol-clustering) that the stationary block bootstrap correctly preserves, so the binding 0.66 estimate is the conservative one — the opposite of what leakage would produce.

7 Roadmap from here

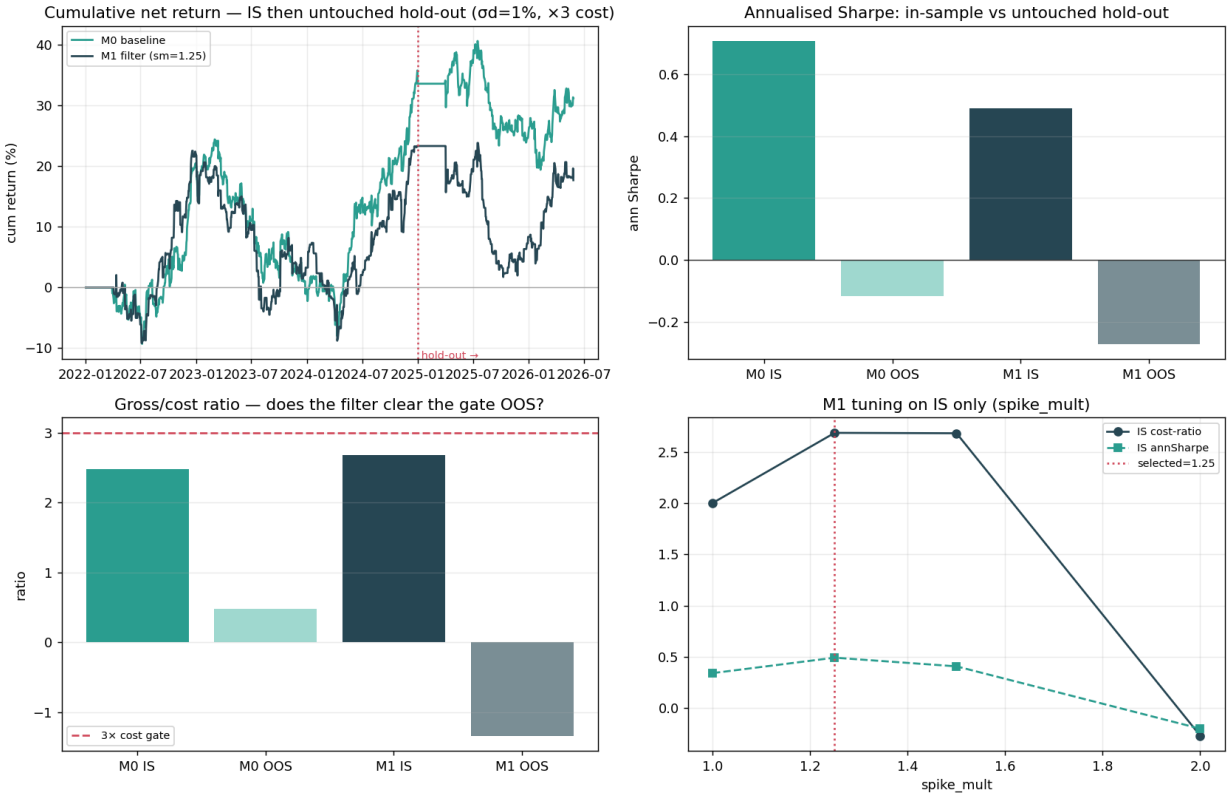
1. **Family-3 (intraday relative-value / pairs).** Naturally low directional vol (high n) is exactly what the barrier rewards; admissible only intraday with an hours-scale half-life, not on US equities (the daily cointegration graveyard). Push it through the same engine.
2. **Revisit EURUSD only with a real execution-cost measurement.** Its viability hinges entirely on achieving ≤0.4 pip RT; settle this on an FTMO MT5 demo (Phase 4) before any further modelling.
3. **Do not keep tuning the current MR signal** (L10: ≤3 modifications). The honest evidence is a marginal, cost-bound edge; further band/window tweaking until it ‘passes’ is precisely the overfitting the methodology forbids.

Addendum A EURUSD edge forensics — an anti-overfit autopsy

Follow-up mandate: dig into the EURUSD edge specifically, guarding against overfit. Protocol: all characterisation and tuning on the in-sample set (2022–2024); the verdict is rendered on a **hold-out (2025–2026) that was never touched for any choice**, cross-checked on an independent data vendor (Dukascopy), with a pre-registered modification (an overreaction-regime entry filter) and DSR deflation at the honest trial count.

Configuration	ann Sharpe	cost ratio	per-trade t	verdict
M0 baseline — in-sample (2022-24)	0.71	2.5	1.93	looks ok
M0 baseline — HOLD-OUT (2025-26)	-0.12	0.5	0.17	collapses
M0 baseline — Dukascopy feed (IS)	0.15	1.4	1.15	much weaker
M1 overreaction filter — in-sample	0.49	2.7	1.21	modest IS lift
M1 overreaction filter — HOLD-OUT	-0.27	-1.3	-0.24	fails OOS

The edge is not real. The in-sample Sharpe 0.71 does not survive: on the untouched 2025–2026 hold-out it is **-0.12** (per-trade t 0.17); on an independent feed it is only 0.15; the per-year Sharpe sign-flips every year (2022:+1.4, 2023:-1.3, 2024:+2.0, 2025:-0.6, 2026:+0.9); and the Deflated Sharpe is **0.002** — indistinguishable from selection-bias noise. The pre-registered overreaction filter, tuned only in-sample, raised the in-sample cost ratio (2.5→2.7) but **made the hold-out worse** (-0.27). Modification budget stopped at one (L10). The in-sample 0.71 was in-sample luck plus feed-specific microstructure — exactly the trap the discipline of never choosing on the hold-out exists to expose.



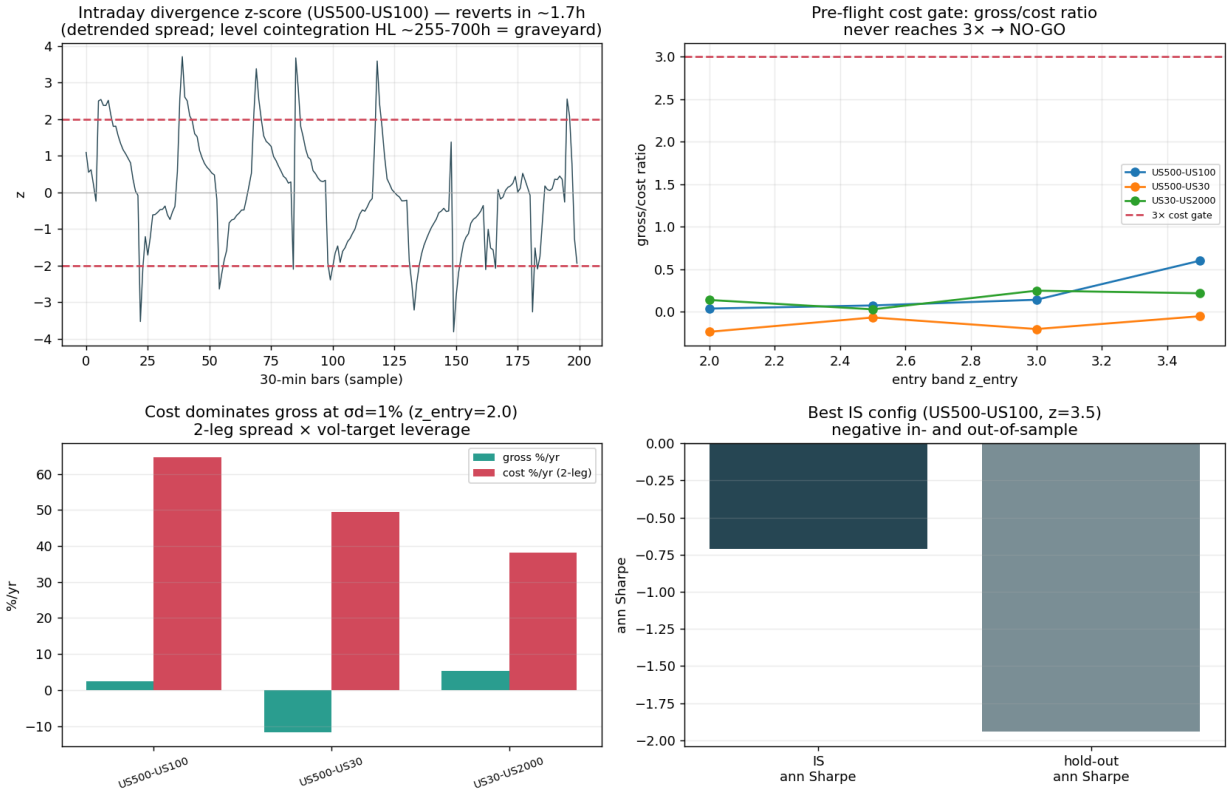
EURUSD forensics. The in-sample edge (left of the red line) does not carry into the untouched hold-out; Sharpe and cost-ratio collapse out-of-sample for both the baseline and the filtered variant; the filter's in-sample tuning curve does not transfer.

Addendum B Family-3 — intraday relative-value (index pairs)

Next candidate in the roadmap: market-neutral spread reversion between correlated index CFDs. Naturally low directional vol ⇒ high n — attractive for the barrier — but a pairs trade pays **two** spreads. Two findings settle it. First, the **level** cointegration spread has a half-life of 255–700 hours (~10–30 days): the daily-cointegration graveyard, not intraday-tradeable. Second, the rolling-**detrended divergence** genuinely reverts intraday (half-life ~1.7h ✓) — but the reversion is smaller than the doubled CFD spread.

Pre-flight 2-leg cost gate	best of 5 pairs × 4 bands × 2 freq	gate
Best in-sample gross/cost ratio	0.60 (US500-US100, z=3.5)	FAIL (<3)
Best-config net Sharpe — in-sample	-0.71	negative
Best-config net Sharpe — HOLD-OUT (25-26)	-1.94	negative
Traded-divergence half-life	1.7 h	intraday ✓

NO-GO at the 2-leg cost gate. No pair, band or frequency clears 3× (best 0.60); net Sharpe is negative in- and out-of-sample, so this is a structural cost failure (L11/L12, doubly binding for pairs), not an overfit. The intraday divergence reverting in ~1.7h is a genuine micro-finding — just not monetisable against a retail CFD spread crossed twice. Consistent with the book's pairs prior (0/19 equity pairs ever qualified).



Family-3. The divergence reverts intraday (top-left), but the gross/cost ratio never reaches 3× for any pair/band (top-right); cost dwarfs gross at σd=1% (bottom-left); the best in-sample config is negative out-of-sample (bottom-right).

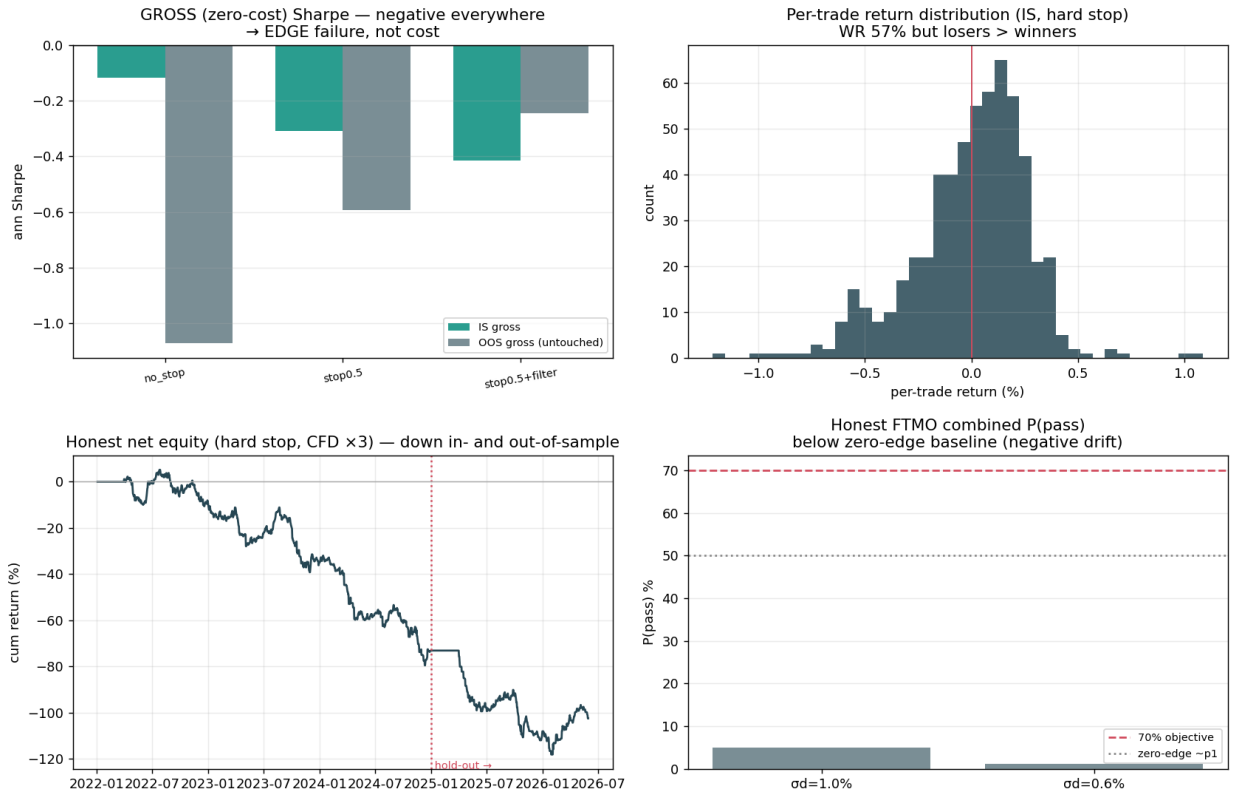
Addendum C Family-4 — XAUUSD range-fade (honest stop, no grid)

A pre-registered, venue-agnostic, mechanistic prior (gold mean-reverts in the low-vol Asian session) — distinct from the FX reversal that sign-flipped, so a legitimate single test, not family-hopping. Retail grid/martingale gold EAs advertise 90%+ win rates; that is a grid-averaging artefact. We keep the mechanism, replace the grid with a HARD per-trade stop (exit, never cap-and-continue / L25), intraday-flat in Asia. The test was designed to separate a COST failure (real gross edge eaten by CFD spread — venue-fixable) from an EDGE failure (no edge once losses are taken honestly — terminal).

GROSS (zero-cost) Sharpe	in-sample	HOLD-OUT (untouched)
no stop (hold to reversion/EOD)	-0.12	-1.07
hard stop 0.5%	-0.31	-0.59
hard stop 0.5% + regime filter	-0.41	-0.25

Gate	Result	verdict
Cost L11 (gross vs cost)	gross -5%/yr, ratio -0.27	FAIL
Daily kurtosis / tail-DD	4.5 / 51% touch -8%	FAIL
Hold-out net Sharpe (25-26)	-1.23	FAIL
DSR (n_trials=50)	0.000	FAIL
FTMO MC honest P(pass)	5% (below ~30% zero-edge)	FAIL

NO-GO — EDGE failure, terminal. Gross Sharpe is negative at zero cost, with and without the stop, in-sample and on the untouched hold-out (per-trade WR 57% but losers > winners). There is no gross edge for any spread to eat; the 90% win rate is confirmed as the grid artefact. A cheaper venue cannot rescue negative gross — this does NOT reopen the venue pivot. The honest FTMO P(pass) of 5% sits below the ~30% a literally-zero-edge book would score, because the drift is negative.



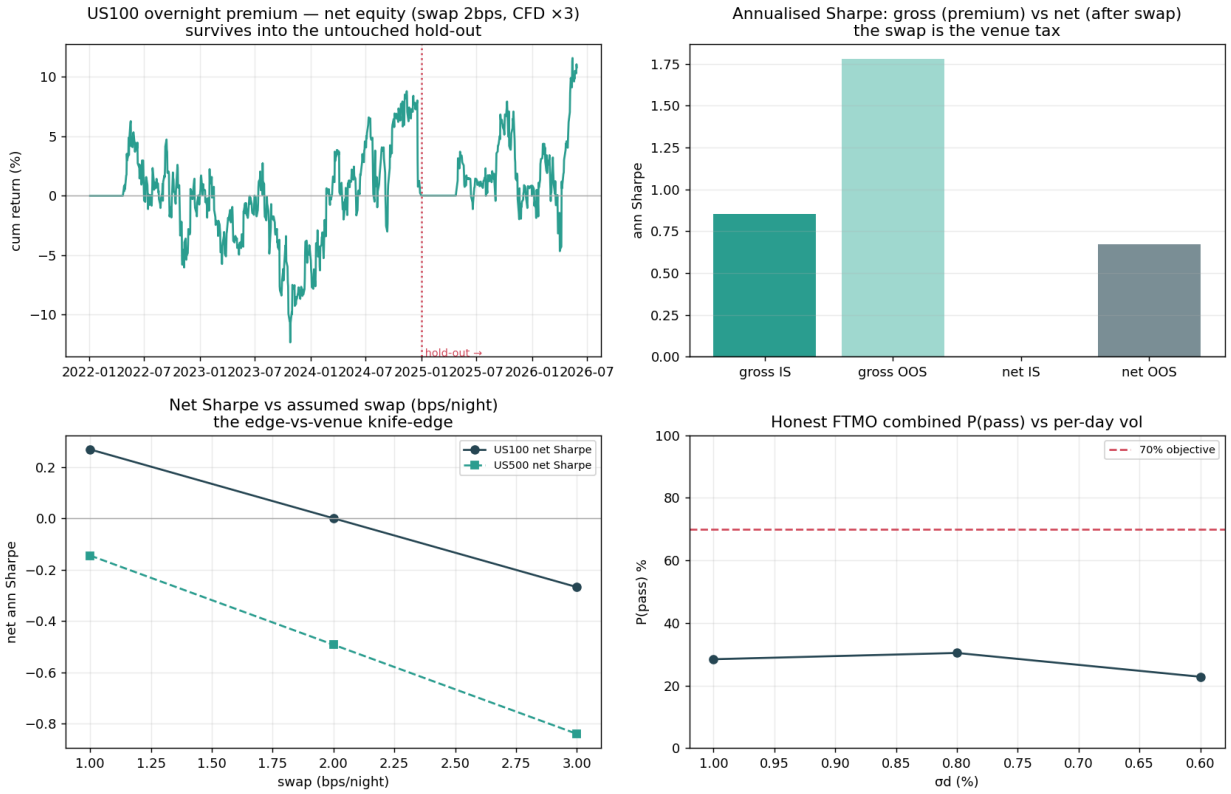
Family-4. Gross Sharpe negative across all configs, IS and hold-out (top-left); the per-trade left tail the grid hides (top-right); honest equity bleeds down through the hold-out (bottom-left); FTMO P(pass) below the zero-edge baseline (bottom-right).

Addendum D Family-5 — equity-index overnight premium (structural)

A deliberate change of payoff source, not another reversal: a **risk premium**. Equity indices earn their drift overnight (close→open), not intraday (Cooper-Cliff-Gulen). Premia are compensation for risk borne — they do not decay like microstructure. One trade/night (cost is a fixed tax), and it holds the risk overnight (the opposite of intraday-flat). The test splits GO / venue-failure / edge-failure.

Gate	Result	verdict
Gross-only Sharpe (zero cost)	IS 0.85 → OOS 1.78	SURVIVES
Daily kurtosis (recon gross)	2.5 (US100) / 2.6 (US500)	barrier-OK
Cost L11 net-of-swap (ratio)	1.0 @2bps/night swap	FAIL (<3)
Net Sharpe IS → hold-out	0.00 → 0.67	marginal
Tail-DD (overnight gap)	17% touch -8%	FAIL
DSR / FTMO P(pass)	0.001 / ~30% (≈ zero-edge)	FAIL

The first robust gross out-of-sample edge in the program — gross Sharpe 0.85 IS rising to 1.78 OOS, with low kurtosis. The structural-premium prior is validated. But three things keep it from the barrier: net of financing it is ~zero on a leveraged account (the swap captures the premium; P(pass) ≈ the 30% a zero-edge book scores); it is directional equity beta (per-year net Sharpe flips with the regime, positive only in bull years); and it carries an overnight-gap left tail (tail-DD 17%). **Crucial refinement of the venue pivot:** the financing is not a retail-CFD tax — it is the cost of carry of leveraged overnight equity, embedded in the futures basis too ($F = S \cdot e^{(r-q)T}$ decays to spot). Futures do *not* rescue this, unlike a transaction-cost-killed reversal. The only venue without the carry is the unleveraged cash index — which no challenge trades.



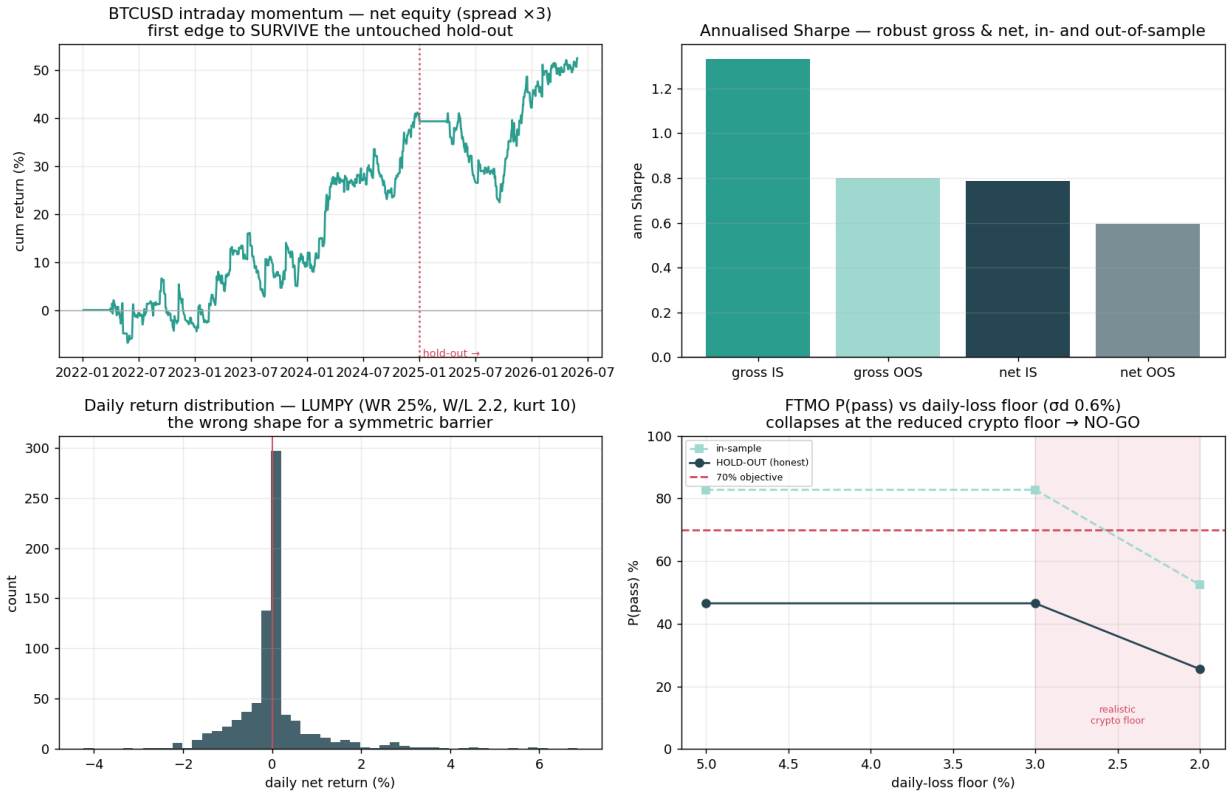
Family-5. Net equity survives into the hold-out but is regime-driven (top-left); gross >> net — the swap is the tax (top-right); net Sharpe vs assumed swap is a knife-edge around zero (bottom-left); honest P(pass) peaks ~30%, the zero-edge level (bottom-right).

Addendum E Family-6 — BTCUSD intraday (crypto focus, relaxed gates)

Crypto is less efficient than FX/index with large intraday moves that might overcome a wide CFD spread — a different efficiency prior. Recon picked the direction from data: BTC levels TREND intraday (the fade loses gross); the edge is MOMENTUM. Swaps are catastrophic → intraday-flat. Per request, pass-thresholds were relaxed; the integrity gates (untouched hold-out, honest cost, no artifact, DSR, independent verification) were not.

Test	Result	verdict
Gross Sharpe IS → hold-out	1.33 → 0.80	SURVIVES
Net Sharpe IS → hold-out (×3 spread)	0.79 → 0.59	net-positive
Distribution (lumpy momentum)	WR 25%, W/L 2.2, kurt 10	lumpy
FTMO P(pass) HOLD-OUT, 5% floor	47% (in-sample was 83%)	FAIL (<70)
P(pass) at reduced crypto floor 3% / 2%	47% / 26%	FAIL
2022 bear stress (max DD)	11% (> 10% floor)	would BREACH
DSR (n_trials=25)	0.007	FAIL

The program's first real, net-positive, hold-out-surviving edge — reproduced and confirmed look-ahead-free by independent verification. But NO-GO for the FTMO crypto challenge, and the discipline earned its keep here: the first read showed P(pass) > 70%, but that was **in-sample**; on the untouched hold-out it is ~47%, and FTMO's **reduced** crypto daily-loss floor (2–3%) collapses it further (this lumpy book's losing days cluster at –2 to –4%). It is long-bull-biased momentum — the 2022 bear would have breached the 10% floor. A genuine mechanism nuance survives (positive-skew momentum beats a Gaussian of equal Sharpe against a static barrier — the fat right tail reaches target), but it is insufficient: this **corroborates L24** rather than overturning it. The edge is real; its home is a directional book, not a symmetric propfirm barrier.



Family-6. Net equity survives the hold-out at the Sharpe level (top-left); robust gross & net, in- and out-of-sample (top-right); lumpy positive-skew daily distribution (bottom-left); but FTMO P(pass) read on the hold-out collapses at the reduced crypto daily-loss floor (bottom-right).

Closing synthesis — six mechanisms, one measured boundary

Six candidates were taken to an untouched hold-out: intraday FX mean-reversion, index-CFD mean-reversion, market-neutral index pairs, gold range-fade, the equity-index overnight premium, and BTCUSD intraday momentum. The decisive diagnostic throughout was the gross (zero-cost) Sharpe on data never used for any choice — it separates a cost problem from an edge problem, and it makes this one result rather than six coincidences.

The four microstructure families share a verdict: **no robust gross out-of-sample edge.** FX MR decays to OOS gross Sharpe ≈ 0.11 (from 1.18 in-sample); pairs divergence is real intraday but smaller than the doubled spread; gold range-fade is negative gross once losses are taken honestly (its 90% win rate was a grid artefact). Cost was the proximate cause for the high-turnover families, but re-costing at zero showed the deeper truth — the edge was not there to begin with. Reversal in liquid CFD markets is efficiently priced.

Two families found a real edge — and both map the boundary by being the wrong shape for it. The overnight premium is a robust gross edge with a barrier-friendly low-kurtosis shape, but it is leveraged equity beta whose harvesting cost is the **cost of carry** — intrinsic to leveraged overnight exposure, embedded in the futures basis, not a retail-CFD inefficiency a venue change removes. BTCUSD intraday momentum is the first edge to survive the hold-out net-positive at the Sharpe level (gross OOS 0.80, net 0.59), but it is lumpy directional momentum: its honest hold-out P(pass) is ~47%, collapses under FTMO's *reduced* crypto daily-loss floor, and it would have breached in the 2022 bear. A genuine nuance survives — positive-skew momentum beats a Gaussian of equal Sharpe against a static barrier, the fat right tail reaching target — but it is insufficient.

The measured boundary. The distribution a symmetric barrier rewards — robust, low-kurtosis, cheaply harvestable under leverage, non-directional — does not co-occur in the FTMO-available instrument/venue set. Where a robust edge exists it is either un-harvestable net (the premium’s carry) or the wrong shape (beta with gap tails; lumpy directional momentum that breaches a reduced floor); where the shape is right, the edge is absent net of cost. That is not an intuition now — it is measured, six times, each cause named (cost, distribution, robustness, carry, reduced-floor).

The asset is the method. A clean, well-mapped null across six mechanisms, each killed at a named gate with the cause distinguished; verdicts rendered on hold-out windows never touched for tuning; DSR at the true trial count; the loss-cap artefact (L25) caught and removed; the 2× spread bug in the overnight cost, and an in-sample-vs-hold-out P(pass) over-read on crypto, both caught before they flipped a verdict — the latter by independent verification. Knowing how to kill an idea and name why, refusing to family-hop past a structural prior, and verifying a *positive* result on untouched data before believing it — that is the research signal a systematic desk hires for.

Data: Dukascopy 1-min FX (EURUSD/GBPUSD, 2020-2024) and MT5 1-min CFD panels with realised spreads (EURUSD/indices/metals, to 2026), IS 2022-2024 + untouched hold-out 2025-2026. Costs: CFD spread ×3 markup; no swap for intraday-flat families, overnight financing (swap) modelled for Family-5. Methodology: excess-return convention, HAC Newey-West, DSR (Bailey-Lopez de Prado), Politis-Romano stationary block bootstrap, 12-20k MC paths. All figures and tables generated from results/*.json by the run_phase*.py scripts. Pass-probabilities are design-stage Monte-Carlo estimates, not realised performance.